1. What is the formal academic equivalent of the phrasal verb 'carry out'?

2. What is the formal Latinate equivalent of 'find out' in research?

3. What is the formal academic replacement for 'look into'?

4. What is the formal equivalent of 'break down' when referring to macromolecules?

5. What is the formal academic verb replacing 'point out'?

6. What is the formal verb for 'set up' in laboratory equipment contexts?

7. In biochemical terminology, the suffix '-ase' denotes an:

8. The prefix 'thermo-' in thermophilic organism means:

9. The Greek root 'cyte' or 'cyto-' refers to:

10. The biological suffix '-lysis' indicates:

11. What does the root 'gen' signify in 'genetics' and 'pathogen'?

12. Which phrasal verb means 'to eliminate or exclude from consideration'?

13. What is the formal equivalent of 'give off' in gas production?

14. Choose the correct word: 'High temperature had an adverse ___ on enzyme stability.'

15. Choose the correct word: 'The mutation will ___ the binding affinity of the receptor.'

16. The term 'in vitro' literally translates from Latin as:

17. The term 'in vivo' refers to biological processes occurring:

18. The modern term 'in silico' denotes research conducted:

19. Which word means 'complete destruction of all viable microorganisms including spores'?

20. The term 'accuracy' in laboratory measurements refers to:

21. The term 'precision' in laboratory measurements refers to:

22. What is the formal verb replacing 'bring about' in experimental outcomes?

23. What is the formal equivalent of 'put forward' regarding hypotheses?

24. What does the biological prefix 'auto-' denote?

25. The prefix 'hetero-' means:

26. The prefix 'homo-' means:

27. Which phrasal verb means 'to support with empirical data or evidence'?

28. What is the formal Latinate equivalent of 'back up'?

29. Choose the correct word: 'The two research groups formed a ___ partnership.'

30. Why is 'complementary' correct in question 29?

31. A 'complimentary' reagent in a promotional vendor packet means it is:

32. What is the formal academic verb replacing 'turn into'?

33. What is the formal verb replacing 'call off' regarding an experiment?

34. Choose the correct word: 'The data consisted of twelve ___ numerical measurements.'

35. The word 'discrete' means:

36. The word 'discreet' means:

37. In genetics, the totality of all genetic material in an organism is the:

38. The suffix '-ome' in modern biotechnology refers to:

39. What does the prefix 'poly-' signify in 'polymerase'?

40. What is the formal equivalent of 'speed up' in enzymatic reactions?

41. Identify the sentence that replaces informal phrasal verbs with formal academic vocabulary:

42. Select the correct academic collocation: 'The empirical data ___ the initial hypothesis.'

43. Why do careful scientific writers prefer 'supports the hypothesis' over 'proves the hypothesis'?

44. Choose the correct confusable word: 'The buffer was chosen based on the underlying chemical ___.'

45. In 'Dr. Rao is the ___ Investigator of the CRISPR project', the correct word is:

46. Identify the separable phrasal verb that can take an object between verb and particle:

47. Which phrasal verb CANNOT be separated by an object (*look it into is ungrammatical)?

48. Choose the correct word: 'The researchers attempted to ___ a clear antibody response in rabbits.'

49. The word 'illicit' means:

50. Choose the correct word: 'The elusive protein managed to ___ isolation for over two decades.'

51. The verb 'allude' means to:

52. Select the correct academic collocation: 'We must ___ a comprehensive risk assessment.'

53. Choose the correct word: 'Before the procedure, centrifugation must ___ chromatography.'

54. The verb 'precede' means to:

55. The verb 'proceed' means to:

56. Identify the appropriate adjective-noun collocation in biostatistics:

57. What is the difference between 'continual' and 'continuous' in laboratory monitoring?

58. Choose the correct word: 'The bioreactor maintained ___ agitation for 72 hours without pause.'

59. Choose the correct word: 'The alarm sounded ___ throughout the morning, every twenty minutes.'

60. What is the formal academic verb replacing 'wipe out' regarding microbial pathogens?

61. What does the Latin prefix 'retro-' mean in 'retrovirus' and 'retrograde'?

62. What does the Greek prefix 'endo-' signify in 'endocytosis' and 'endonuclease'?

63. What does the prefix 'exo-' mean in 'exocytosis' and 'exonuclease'?

64. Choose the correct phrasal verb: 'The bio-technician ___ an automated script to parse the FASTA file.'

65. What is the formal academic verb replacing 'came up with'?

66. Select the sentence with correct technical verb-preposition collocation:

67. Choose the correct collocation: 'The clinical findings ___ serious ethical concerns.'

68. What is the formal verb replacing 'get rid of' in hazardous chemical disposal?

69. Choose the correct word: 'The patient showed high ___ to penicillin.'

70. The word 'sensibility' in communicative English refers to:

71. What is the biological meaning of the portmanteau word 'bioinformatics'?
computational data science) (B) Biology + Information desk

72. Identify the word formed by compounding in molecular biology:

73. What is the formal Latinate verb replacing 'narrow down' in identifying candidate genes?

74. Select the correct collocation: 'We reached a definitive ___ regarding the mechanism of action.'

75. Choose the correct word: 'The clinical efficacy of the vaccine was demonstrated beyond ___.'

76. Identify the sentence that violates formal academic register by using colloquial phrasal verbs:

77. Select the sentence showing impeccable deployment of high-register academic vocabulary:

78. Choose the correct word: 'The chemical assay yielded ___ results that could not be replicated.'

79. The word 'spurious' in experimental analysis specifically denotes findings that are:

80. Choose the correct word: 'The investigator made a subtle ___ to Mendel's seminal 1866 paper.'

81. In bioprocess kinetics, what is the exact semantic distinction between 'quiescent' and 'senescent' cells?

82. Select the sentence with correct usage of the high-register transition 'notwithstanding':

83. What does the Latinate term 'mutatis mutandis' mean in technical comparative protocols?

84. Choose the correct word: 'The novel synthetic scaffold is completely ___ to enzymatic degradation.'

85. In biomedical science, a substance or disease that is 'refractory' is:

86. Identify the sentence that uses 'mitigate' correctly in laboratory risk management:

87. Why is 'mitigates against' considered an uneducated error in academic editing?
alleviate) with intransitive 'militate against' (work against)   (B) 'Mitigate' requires a preposition

88. Choose the correct word: 'The newly discovered retro-transposon has an ___ distribution across the genome.'

89. What is the precise morphological meaning of the term 'epigenetics'?
upon/above) + genetics (heritable code)   (B) Epi (inside) + genetics
without) + genetics   (D) Epi (before) + genetics

90. Select the formal academic verb that best replaces 'stand up to' in: 'The model can stand up to scrutiny':

91. Choose the correct word: 'The researcher was ___ to draw definitive conclusions from unverified data.'

92. The adjective 'loath' means:

93. The verb 'loathe' means:

94. Identify the correct technical word for a chemical compound that accelerates a reaction without being consumed:

95. What is the formal academic verb replacing 'phase out' in phasing out mercury reagents?

96. Choose the correct word: 'The presence of the 500 bp band is ___ of successful recombinant ligation.'

97. Select the sentence with flawless academic collocations and formal Latinate vocabulary:

98. Choose the correct word: 'The therapeutic dose must be calibrated so as not to ___ toxicity.'

99. The verb 'exacerbate' means to:

100. Identify the sentence that represents the pinnacle of communicative English precision, technical vocabulary, and professional academic register:

SECTION B: DESCRIPTIVE & SUBJECTIVE QUESTIONS

Q1. Analyze the morphological taxonomy of Greek and Latin roots, prefixes, and suffixes in biotechnology and life sciences.

Q2. Contrast informal phrasal verbs with formal Latinate verbs in scientific publications, providing a comprehensive translation matrix.

Q3. Examine the syntactic mechanics of separable versus inseparable phrasal verbs in technical writing.

Q4. Analyze precision in biotechnology terminology, distinguishing between 'sterilization', 'disinfection', 'antisepsis', and 'sanitization'.

Q5. Examine the semantic distinction between 'accuracy' and 'precision' in bio-analytical assay validation.

Q6. Analyze confusing word pairs in academic writing: 'Affect' versus 'Effect', and their rare reverse grammatical usages.

Q7. Examine the technical distinction between 'sensitivity' and 'specificity' in biomedical diagnostic tests.

Q8. Detail the semantic and grammatical differences between 'Complement' and 'Compliment' in scientific discourse.

Q9. Analyze word-formation processes in biotechnology: derivation, compounding, blending, and acronymy.

Q10. Examine strong academic collocations with the nouns 'hypothesis', 'experiment', 'result', and 'conclusion'.

Q11. Contrast 'discrete' versus 'discreet' and 'continuous' versus 'continual' in bioprocess engineering.

Q12. Analyze the semantic field of scientific verbs of revelation: 'demonstrate', 'indicate', 'suggest', 'prove', 'imply'.

Q13. Examine the technical distinction between 'in vitro', 'in vivo', 'in situ', and 'in silico' research modalities.

Q14. Analyze the precision of 'precede' versus 'proceed' and 'principal' versus 'principle' in SOPs.

Q15. Examine phrasal verbs with 'break' and their formal replacements in biochemistry and microbiology.

Q16. Analyze the distinction between 'elicit' and 'illicit' and 'allude' and 'elude' in biomedical papers.

Q17. Detail the morphological meaning and diagnostic applications of the 'Omics' cascade in biotechnology.

Q18. Analyze phrasal verbs with 'set' in laboratory instrumentation and their formal replacements.

Q19. Examine confusing words: 'Eradicate', 'Eliminate', 'Ablate', and 'Extirpate' in cellular and genetic targeting.

Q20. Analyze transitional discourse markers and signposting vocabulary in peer-reviewed scientific arguments.

Q21. Examine the semantic distinction between 'mitigate' and 'militate' and common misuse in biotechnology risks.

Q22. Analyze phrasal verbs with 'carry' ('carry on', 'carry out', 'carry over') in laboratory protocols.

Q23. Examine the technical distinction between 'homogeneous' and 'heterogeneous' across biochemical phases.

Q24. Analyze Latin abbreviations in scientific text: 'i.e.' versus 'e.g.', and 'in situ' versus 'ex situ' conservation.

Q25. Examine phrasal verbs with 'give' ('give up', 'give off', 'give in', 'give rise to') in biological descriptions.

Q26. Analyze the distinction between 'presumptive', 'putative', and 'proven' biological markers.

Q27. Examine confusable verbs: 'Assay', 'Appraise', 'Analyze', and 'Ascertain' in laboratory documentation.

Q28. Analyze phrasal verbs with 'bring' ('bring about', 'bring down', 'bring up') in research communications.

Q29. Examine precision in technical quantitative adjectives: 'negligible', 'marginal', 'appreciable', 'substantial'.

Q30. Synthesize strategies for building, mastering, and retaining advanced academic vocabulary in biotechnology.

SECTION C: ANSWER KEY - MULTIPLE CHOICE QUESTIONS

Q1: [A]
Q2: [A]
Q3: [A]
Q4: [B]
Q5: [A]
Q6: [A]
Q7: [B]
Q8: [B]
Q9: [C]
Q10: [B]
Q11: [A]
Q12: [A]
Q13: [B]
Q14: [B]
Q15: [B]
Q16: [B]
Q17: [A]
Q18: [A]
Q19: [B]
Q20: [A]
Q21: [B]
Q22: [A]
Q23: [A]
Q24: [B]
Q25: [B]
Q26: [B]
Q27: [A]
Q28: [A]
Q29: [A]
Q30: [A]
Q31: [B]
Q32: [A]
Q33: [A]
Q34: [A]
Q35: [B]
Q36: [B]
Q37: [A]
Q38: [B]
Q39: [B]
Q40: [A]
Q41: [B]
Q42: [B]
Q43: [A]
Q44: [B]
Q45: [B]
Q46: [B]
Q47: [B]
Q48: [B]
Q49: [B]
Q50: [B]
Q51: [B]
Q52: [C]
Q53: [B]
Q54: [A]
Q55: [A]
Q56: [B]
Q57: [A]
Q58: [A]
Q59: [B]
Q60: [A]
Q61: [B]
Q62: [B]
Q63: [B]
Q64: [A]
Q65: [A]
Q66: [B]
Q67: [B]
Q68: [A]
Q69: [A]
Q70: [B]
Q71: [A]
Q72: [A]
Q73: [A]
Q74: [B]
Q75: [B]
Q76: [B]
Q77: [A]
Q78: [B]
Q79: [B]
Q80: [B]
Q81: [A]
Q82: [C]
Q83: [A]
Q84: [A]
Q85: [B]
Q86: [A]
Q87: [A]
Q88: [A]
Q89: [A]
Q90: [D]
Q91: [A]
Q92: [A]
Q93: [A]
Q94: [B]
Q95: [A]
Q96: [A]
Q97: [A]
Q98: [A]
Q99: [A]
Q100: [A]

MCQ Scoring Guide & Difficulty Distribution:

Questions 1 to 40: Foundational & Conceptual [Easy] | Questions 41 to 75: Analytical & Applied [Moderate]

Questions 76 to 100: Complex Numerical & Advanced Problem Solving [Difficult]

[illegible]

- "wipe out" -> eradicate, eliminate, obliterate
- "cut down on" -> reduce, attenuate, diminish
- "get rid of" -> dispose of, discard, neutralize

3. Stylistic Rule:
In Title, Abstract, Introduction, and Results sections of research manuscripts, systematically supplant multi-word phrasal verbs with their formal Latinate counterparts.

>> [illegible] 'antisepsis', and 'sanitization'.

- The destruction or chemical elimination of pathogenic microorganisms on INANIMATE surfaces (benches, hoods, floors):
- Critical Distinction: Disinfection destroys vegetative bacteria but does NOT reliably destroy resistant bacterial endospores (e.g., Bacillus or Clostridium spores).
- Agents: 70% ethanol, 10% sodium hypochlorite (bleach), quaternary ammonium compounds.
- Antisepsis:
- The application of non-toxic chemical agents directly to LIVING BIOLOGICAL TISSUE (skin, mucous membranes) to inhibit or arrest microbial proliferation:
- Agents: Chlorhexidine gluconate, povidone-iodine, 70% isopropanol skin wipes.
- Sanitization:
- The reduction of microbial populations on surfaces to safe public health regulatory thresholds:
- Context: Industrial bioprocessing food-grade washdowns.

[illegible]

- Affect (Almost Always a Verb): To exert an influence upon, alter, or produce a change in something:
- "Substrate concentration affects the initial reaction velocity."
- Effect (Almost Always a Noun): The concrete outcome, result, or consequence brought about by a cause:
- "The allosteric inhibitor had a profound effect on enzyme kinetics."

2. Rare Reverse Grammatical Usages in High-Register Prose:
- Effect as a Verb (Meaning 'to bring about / to cause to exist / to execute'):
- Used in formal institutional, managerial, or biochemical contexts:
- "The gene therapy effected a dramatic transformation in patient survival."
- "The committee effected major policy revisions." (Notice: means brought into reality).
- Affect as a Noun (Pronounced with initial stress /'ae.fekt/):
- Strictly a clinical psychiatry / psychology term denoting external emotional expression:
- "The clinical trial participant exhibited a flat affect during neurological evaluation."

3. Mnemonic Diagnostic:
Remember RAVEN: Remember Affect is a Verb, Effect is a Noun.

>>Solution to Question 8: [illegible] scientific discourse.

2. Detailed Definitions:
- Complement (with 'e' - To complete or enhance mutual functionality):
- As a Verb: To supply what is lacking to form a harmonious, integrated whole:
- "The RNA-seq transcriptomic data complement our proteomic mass spectrometry findings."
- As a Noun: An entity that completes another:
- "DNA base pairs form complementary strands: adenine is the complement of thymine."
- In Immunology: The 'Complement System' (a cascade of serum proteases enhancing immune clearance).
- Compliment (with 'i' - Praise or free courtesy):
- As a Verb: To express praise, admiration, or congratulation:
- "The keynote speaker complimented the graduate students on their rigorous poster presentation."
- As an Adjective (Complimentary): Given free of charge as a courtesy:
- "The vendor supplied a complimentary sample of DNA ladder."

3. Diagnostic Memory Hook:
ComplEment with an 'E' adds what is nEcessary to complEte.

>> Solution to Question 30:
[Problem Statement: [illegible] academic collocations [illegible] with the nouns 'hypothesis', 'experiment', 'result', and 'conclusion'.

partnerships (collocations) rather than translating literal synonyms:

2. Collocation Matrix for Core Academic Heads:
- Collocations with 'Hypothesis':
- Verbs: formulate a hypothesis, test a hypothesis, substantiate / corroborate a hypothesis, refute / falsify a hypothesis, advance a hypothesis.
- Adjectives: null hypothesis, plausible hypothesis, working hypothesis, substantiated hypothesis.
- Collocations with 'Experiment':
- Verbs: conduct an experiment, perform an experiment, undertake an experiment, execute an experiment (Avoid *do an experiment).
- Adjectives: controlled experiment, preliminary experiment, benchmark experiment.
- Collocations with 'Results':
- Verbs: results indicate, results corroborate, results suggest, results demonstrate, yield results (Avoid *make results).
- Adjectives: reproducible results, statistically significant results, consistent results, anomalous results.
- Collocations with 'Conclusion':
- Verbs: draw a conclusion, reach a conclusion, arrive at a conclusion, deduce a conclusion (Avoid *take a conclusion).
- Adjectives: definitive conclusion, tentative conclusion, overarching conclusion.

>>Solution to Question 12: [illegible]

[Problem Statement]: [illegible] 'demonstrate', 'indicate', 'suggest', 'prove', 'imply'.

- Prove:
- Absolute deductive certainty; completely eliminates alternative explanations.
- Usage in Biology: Rarely appropriate for empirical findings; strictly reserved for mathematical or logical proofs:
- Caution: Avoid "*These data prove our theory".
- Demonstrate:
- High empirical certainty based on direct, undeniable experimental observation:
- "Cryo-EM reconstructions demonstrate that the ribosome undergoes conformational rotation."
- Indicate:
- Clear, strong evidentiary support without claiming absolute mathematical closure:
- "Western blot analyses indicate elevated expression of target kinase."
- Suggest:
- Prudent scientific deduction; acknowledges that while the proposed interpretation is favored, alternative models remain plausible:
- "These binding kinetics suggest that allosteric regulation is operative."
- Imply:
- Logical consequence that follows indirectly from empirical premises without being stated outright.

[illegible]

1. 'Precede' versus 'Proceed':
- Precede (from Latin prae-cedere: to go before in time, order, or spatial hierarchy):
- "Enzymatic cell lysis must precede nucleic acid purification."
- "Table 1 is preceded by a brief methodological introduction."
- Proceed (from Latin pro-cedere: to go forward, continue, or advance):
- "Once the pellet is washed, proceed immediately to the resuspension step."
- "The fermentation proceeded smoothly toward stationary phase."

2. 'Principal' versus 'Principle':
- Principal (Adjective: chief, primary, most important; Noun: head administrator / primary financial capital):
- "The principal investigator directed the CRISPR clinical campaign."
- "The principal limitation of our study was small sample size."
- Principle (Noun: fundamental truth, scientific law, ethical doctrine):
- "The Beer-Lambert Law is a foundational optical principle."
- "Research was conducted in accordance with ethical principles of animal welfare."

>>Solution to Question 16: [illegible]

2. 'Allude' versus 'Elude':
- Allude (Verb: to make an indirect or passing reference):
- "The author alluded to previous unpublished crystallographic data."
- Note: Governs the preposition 'to' (allude to).
- Elude (Verb: to escape detection, avoid capture, or fail to be achieved):
- "A high-resolution crystal structure of the integral membrane transporter had eluded structural biologists for thirty years."
- "The exact mechanism continues to elude researchers."

[illegible]

- Meaning: The beginning of a persistent physiological state.
- Formal Replacement: Commenced, initiated, developed.
- Example: "Cellular senescence commenced after passage 15."
- Set off:
- Meaning: Triggering a catalytic reaction or emergency alarm.
- Formal Replacement: Trigger, initiate, activate.
- Example: "Thermal runaway activated the automated fire suppression system."
- Set back:
- Meaning: Delaying experimental timelines.
- Formal Replacement: Impeded, delayed, hindered.

[illegible]

- Consequently, Therefore, Thus, Hence, Accordingly.
- Function: Demonstrates direct deductive consequence from empirical observations.
- Concessive Markers:
- Although, Albeit, Even so, Despite, While.
- Function: Acknowledges an experimental limitation without invalidating the core hypothesis.
- Exemplification Markers:
- Specifically, For instance, Notably, Particularly.

>> Solution to Question 22: [illegible] biotechnology risks.

[illegible]

gas."
- Give rise to:
- Meaning: Initiating or causing a biological phenotype or structure.
- Formal Replacement: Generate, induce, yield, produce, prompt.
- Example: "Pluripotent embryonic stem cells give rise to cell types from all three embryonic germ layers."
- Give up:
- Meaning: Abandoning a research inquiry or donating an electron.
- Formal Replacement: Surrender, relinquish, donate.
- Example: "The reducing agent donates electrons to the disulfide bonds."

[illegible]

- Ascertain:
- Denotes finding out with definitive certainty through investigation:
- "Further sequencing was performed to ascertain whether the mutation was somatic or germline."

[illegible]

2. Actionable Lifelong Learning Strategies:
- Etymological Root Deconstruction:
- Systematically dissect unfamiliar terminology into Greek/Latin roots, prefixes, and suffixes. Recognizing that 'omnis' means all, 'potens' means powerful, and '-cyte' means cell instantly demystifies 'omnipotent cell' (pluripotent / totipotent).
- Corpus-Based Reading of High-Impact Journals:
- Read peer-reviewed publications (Nature, Cell, Science, Nucleic Acids Research) with an active lexical notebook, highlighting recurrent collocations, signposting transitions, and formal Latinate verbs.
- Eliminating Conversational Crutches:
- Practice converting spoken procedural descriptions into formal prose: consciously replace 'set up' with 'configure', 'look into' with 'investigate', and 'find out' with 'ascertain'.
- Collocation Mapping:
- Never memorize isolated words in isolation; record complete functional noun-verb and adjective-noun frameworks (e.g., "elicit a response", "refractory to treatment", "corroborate a hypothesis").
- Result: Establishes effortless academic fluency, ensuring graduate theses, patent claims, and journal submissions command immediate professional respect.
